Predicting Lithium-Ion Battery End of Life from Early Test Cycles

GEORGIA NETWORK *for* ELECTRIC MOBILITY
UNIVERSITY OF GEORGIA

Levon Lau & Dr. Xianqiao Wang · University of Georgia College of Engineering · July 2026

Why this matters

Full battery life tests can run hundreds to thousands of cycles — slow and costly for fast-charge R&D.

We ask: can we predict **end-of-life (EOL)** from early-cycle data before capacity visibly fades?

- EOL = first cycle below **80%** of initial discharge capacity
- Dataset: MIT-Stanford-Toyota fast-charging cells (Severson et al., 2019)
- **134** commercial cells after cleaning **140** raw test files

Approach

- **140** raw JSON files → **134** unique cells · cell-level **94 / 20 / 20** train/val/test split
- **44** hand-crafted early-cycle features, including Severson-style **ΔV(Q)** voltage curves
- Classical ML (linear, ElasticNet, random forest, **XGBoost**) vs PyTorch **GRU** on per-cycle trajectories
- Ablation: how many of the first **N** cycles are needed? (**N** = 20, 50, 100)

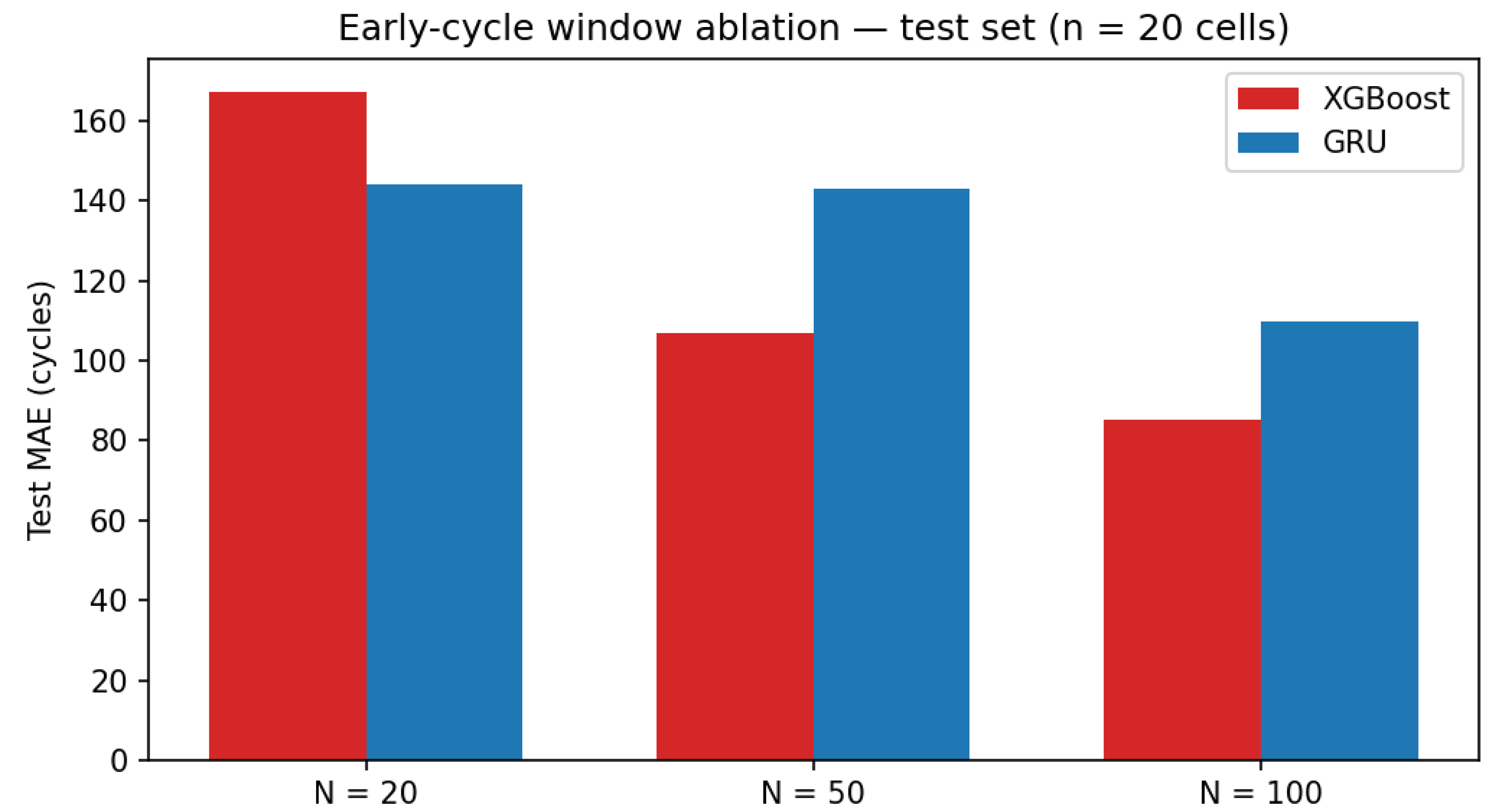
Headline results

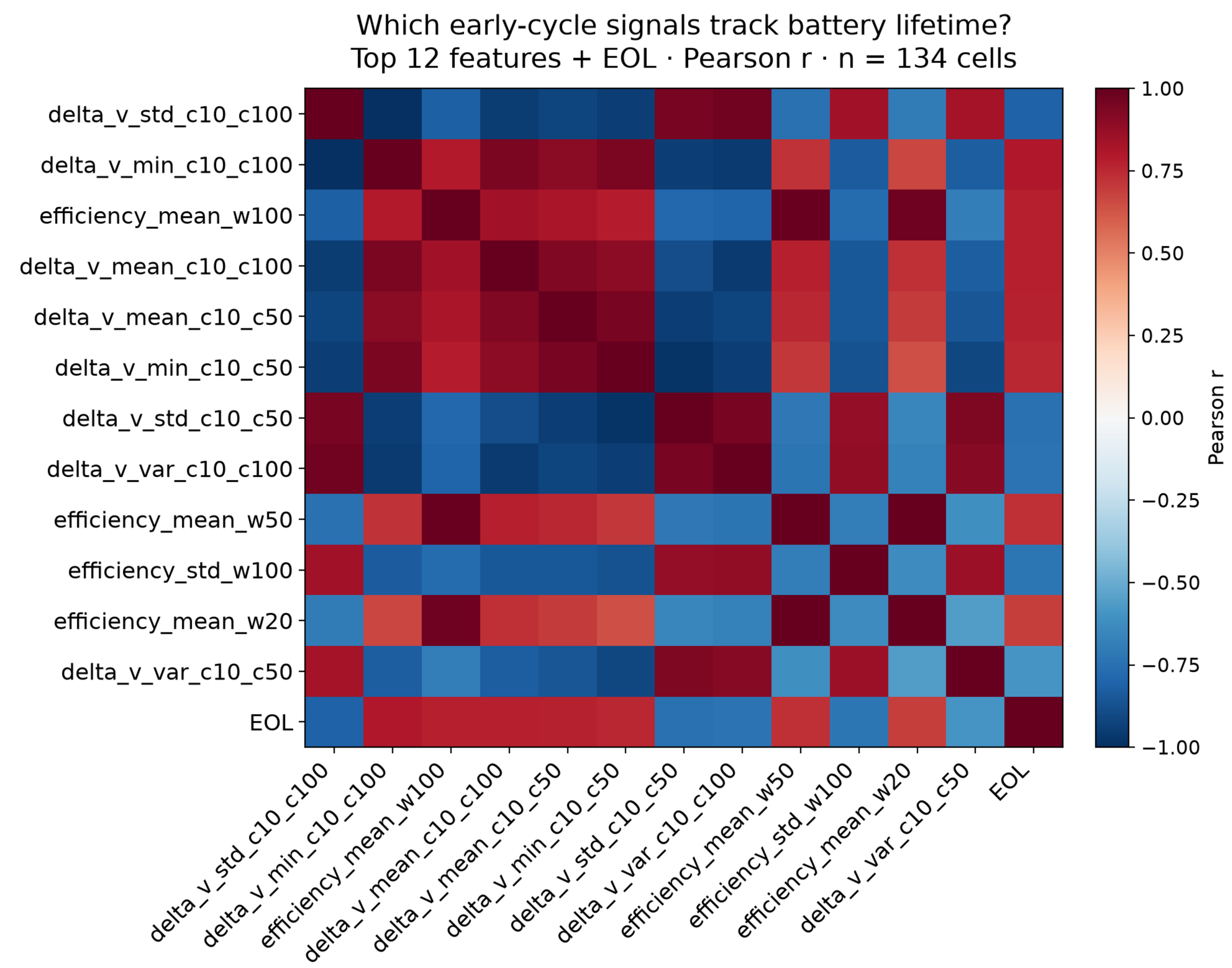
Best holdout model: XGBoost at N = 100 — about 85 cycles test MAE (about 11% MAPE).

- GRU sequence model: about **110** cycles MAE (competitive, not best)
- XGBoost improves sharply with more early data: **167 → 107 → 85** cycles MAE (**N** = 20, 50, 100)
- At **N** = 20 only, GRU beats XGBoost (voltage-curve features unavailable to the tree model)

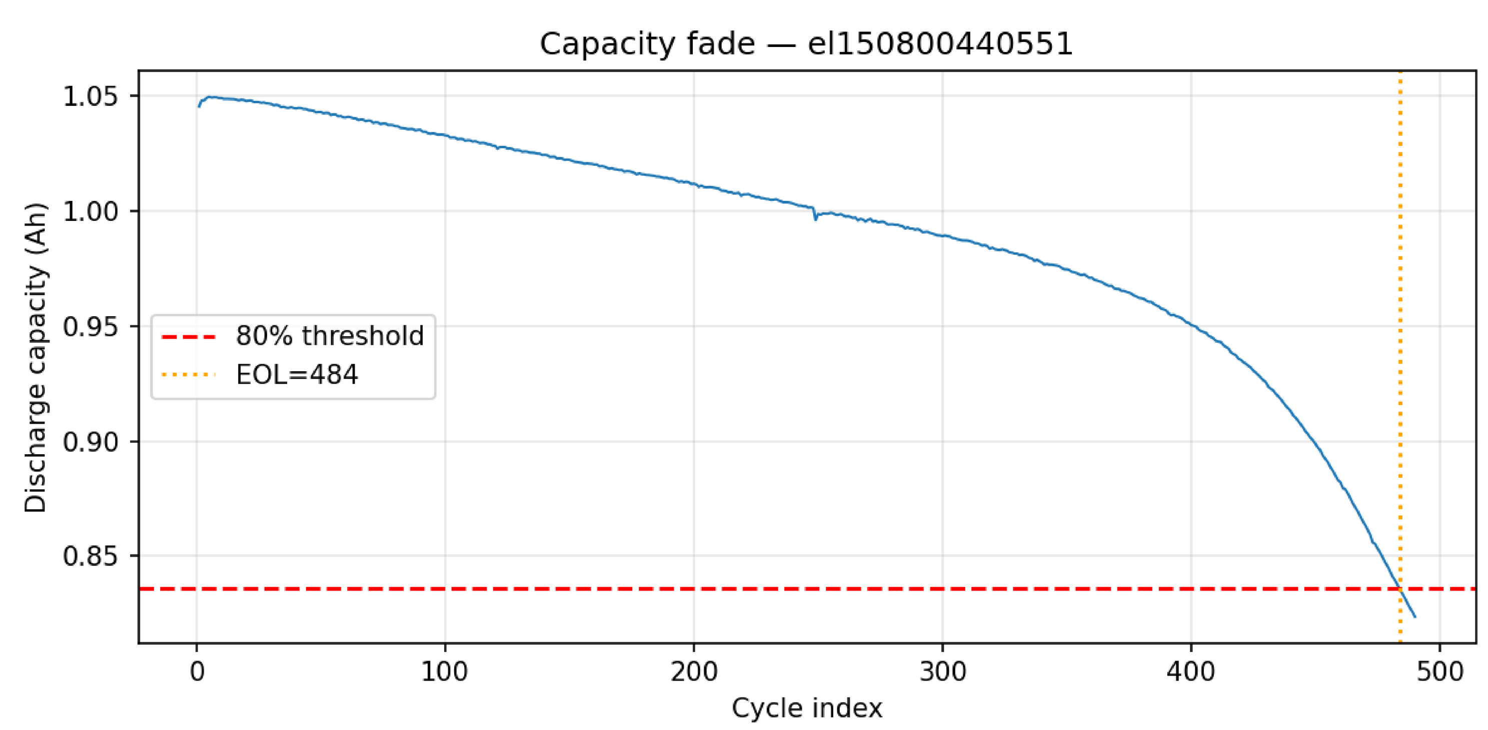
github.com/levonl22/gnem-battery-degradation

How many cycles are needed?





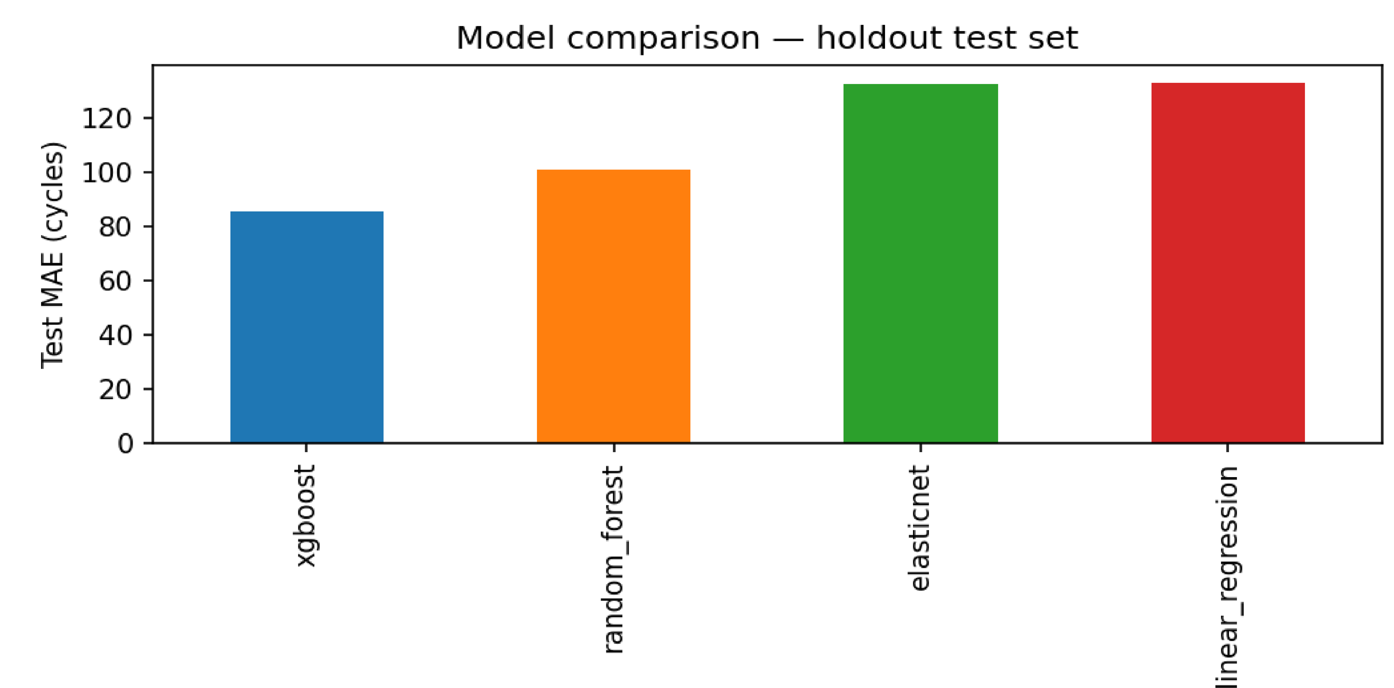
*ΔV(Q) and **energy efficiency** stand out before capacity fade is obvious. Red = longer life; blue = shorter life.



If you only observe the first **N** cycles, how well can you predict end-of-life? We retrained **XGBoost** and a **GRU** at **N = 20, 50, 100** (same 20-cell test holdout).

XGBoost gains the most as **N** grows (**167 → 107 → 85** cycles MAE).

GRU is flatter until **N = 100**. At **N = 20**, GRU wins; at **N = 100**, XGBoost wins — the best model depends on how much early data you have.



Takeaways

Early-cycle measurements can support **fast-charge cell screening** before full degradation is visible.

- About **50–100 cycles** plus voltage-curve features give the strongest tabular signal
- Reproducible open pipeline on GitHub (notebooks **01–10**)

Best overall: XGBoost on 44 features at N = 100.

Limitations

Limitations: 20-cell test holdout; single LFP fast-charge protocol; 134 cells total. Metrics are indicative, not definitive.

Future work: NASA/CALCE validation · ΔV(Q) channels in sequence models · larger cohorts.